Hoang Anh Nguyen Huu

Robotics / Control Engineering Intern

(+84) 365 924 027 | hoanganh94ptit2k5@gmail.com | Hanoi, Vietnam

github.com/HoangAnh301194 | linkedin.com/in/hoanganh94ptit2k5

SUMMARY

Control Engineering and Automation undergraduate with practical experience in ROS 2, mobile robotics, embedded systems, and robot control. Seeking a robotics internship focused on robot software, perception, and motion control.

EDUCATION

Posts and Telecommunications Institute of Technology (PTIT) — Hanoi, Vietnam

B.Eng. Candidate in Control Engineering and Automation — Expected Graduation: 2028

- Specialization: **Robotics and Applied AI**; Current GPA: **3.47/4.0**
- Relevant coursework: Control Theory, Robotics, Embedded Systems

TECHNICAL SKILLS

Programming: C/C++, Python
Embedded Systems: ESP32, Arduino, ESP-IDF, FreeRTOS
Robotics & Control: PID Control; Manipulator Kinematics; ROS2
Computer Vision / AI: OpenCV, YOLO finetuning
Tools: Linux/Ubuntu, Git/GitHub, basic MATLAB/Simulink, PlatformIO, LaTeX

EXPERIENCE

DTT Technology Joint Stock Company (DTT) — 2026 – Present

R&D Intern — Hanoi, Vietnam

- Applied image processing and CNN-based methods for robot recognition and orientation estimation.

Research Laboratory / Academic Research Group — 2024 – Present

Student Researcher — PTIT, Hanoi

- Participated in research activities related to embedded systems, autonomous robots, and applied AI for robotic systems.

SELECTED PROJECTS

Quadcopter Control Prototype | *ESP32, FreeRTOS, PID Control, BLE*

- Built a quadcopter prototype using ESP32, FreeRTOS, PID-based control, and BLE communication for configuration.

Micromouse Maze-Solving Robot | *STM32F411, PID Control, Flood Fill/BFS*

- Developed a micromouse robot using STM32F411, PID-based motion control, and Flood Fill/BFS for maze solving.

6-DOF Robotic Arm Control | *Kinematics, Inverse Kinematics, Trajectory Planning*

- Implemented forward/inverse kinematics and trajectory planning for a 6-DOF robotic arm to generate end-effector paths and joint-space motions.

ADDITIONAL INFORMATION

Languages: Vietnamese (Native), English (Intermediate), Japanese (Intermediate)